

staunton

Chess diagrams, games and tournament tables for Typst

User manual · package version 0.1.0

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1 Introduction

staunton is a Typst package for chess. From a FEN string, a parsed PGN, or a position you build by hand, it produces:

- **boards and diagrams** — captioned, cross-referenceable figures, with labels, highlights, arrows, an optional grid, flexible sizing, custom colours, and bundled SVG piece sets (or a Unicode fallback);
- **games from PGN** — a lazy parser, position navigation by locator (mainline and variations), move play-out, and FEN export;
- **move notation** — localized piece letters, figurine glyphs, NAGs and comments, and diagrams embedded inline;
- **tournament tables** — standings, cross-tables and progress charts from a PGN's results, by player or by team;
- **outlines and references** — diagrams and tables get their own counters and lists;
- **document-wide styling and localization** (six languages, easily extended).

```
#chess-diagram(  
  "r1bqkbnr/pppp1ppp/2n5/4p3/4P3/5N2/  
  PPPP1PPP/RNBQKB1R",  
  caption: [The Ruy Lopez, three moves  
  in.],  
  size: 4cm,  
)
```

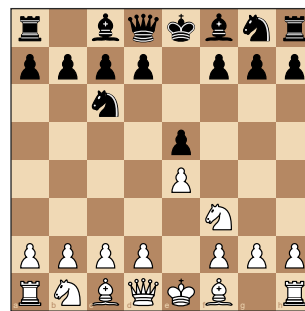


Diagram 1: The Ruy Lopez, three moves in.

This manual is bottom-up. The **board** is the drawing primitive; a **diagram** wraps a board in a referenceable figure; a **position** is the data a board draws; **games** add PGN, notation, and play; then come **tournament tables**, **outlines and references**, and the **document-wide style** settings.

1.1 The name

staunton honours **Howard Staunton** (c. 1810–1874): a leading chess master of his day, organiser of the first international tournament (London, 1851), a chess author and publisher, and the namesake of the standardised **Staunton pattern** chessmen — still the tournament standard.

1.2 Installing and importing

staunton is a Typst package. Import its public API once and every function in this manual is in scope:

```
#import "@preview/staunton:0.1.0": *
```

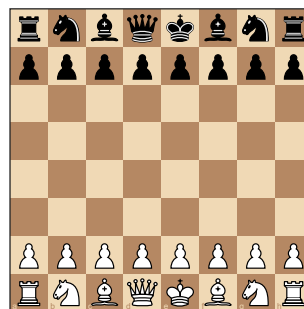
1.3 How to read this manual

In every framed example, the left side is **the code you type** and the right side is **exactly what it renders** — the manual compiles its own examples, so the two can never disagree.

2 The board

`board(source, ...)` draws **just the board** — no caption, no figure — and is the primitive every diagram builds on. `source` is one of: a **FEN string**; a **position** (from `position(...)` or `parse-fen(...)`); or a bare **squares** dict (square name → (kind, color)).

```
#board(
  "rnbqkbnr/pppppppp/8/8/8/PPPPPPPP/
  RNBQKBNR",
  size: 4cm,
)
```



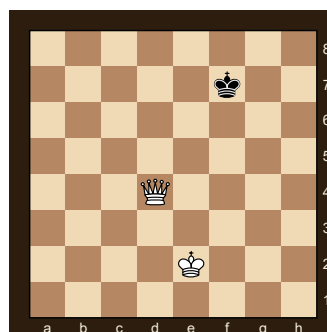
`board` is the variant-agnostic primitive; `chess-board` is the standard-chess sugar over it (same rendering, but it documents the variant and rejects a non-standard source). Use `board` inline in text or inside your own layout; reach for a **diagram** (next chapter) when you want a captioned, referenceable figure.

The rest of this chapter covers the board's drawing options: labels, highlights, arrows, the grid, coordinates, size, colours, orientation, and piece sets — all of which a `chess-diagram` accepts too.

2.1 Labels

`label-mode` chooses how files and ranks are drawn — "on-square" (default, tucked into the corner squares), "outside" (a gutter strip), or "border" (a themed band, styled by `border-theme`). `labels: false` suppresses them.

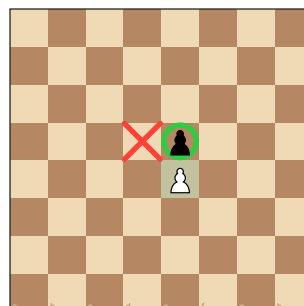
```
#board(
  "8/5k2/8/8/3Q4/8/4K3/8",
  label-mode: "border",
  border-theme: "brown",
  size: 3.8cm,
)
```



2.2 Highlights

`highlight` marks squares; each entry is a square name (drawn with `highlight-shape`, default "filled"), a (square, color) pair, or a dict (square: .., shape: .., color: ..) where shape is "filled", "cross", or "circle". By convention a **cross** marks an empty square.

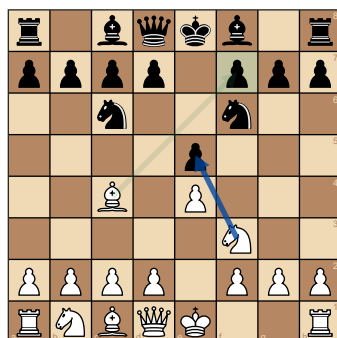
```
#board(
  "8/8/8/4p3/4P3/8/8/8",
  highlight: (
    "e4",
    (square: "e5", shape: "circle"),
    (square: "d5", shape: "cross"),
  ),
  size: 4cm,
)
```



2.3 Arrows and the grid

`arrows` draws arrows; each entry is a (from, to) or (from, to, color) tuple, or a dict (from: .., to: .., color: ..). A missing color uses `arrow-color`. Arrows scale with the board and flip with it. A `grid: true` overlay draws thin lines between the squares.

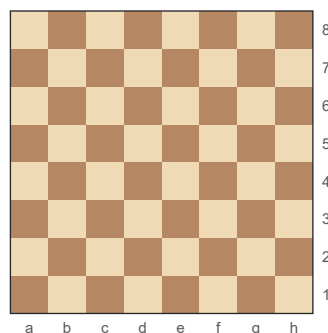
```
#board(
  "r1bqkb1r/pppp1ppp/2n2n2/4p3/2B1P3/5N2/PPPP1PPP/RNBQK2R",
  grid: true,
  highlight: ("f7",),
  arrows: (("c4", "f7"), ("f3", "e5", rgb(0, 70, 160, 200))),
  size: 4.4cm,
)
```



2.4 Coordinates and non-square boards

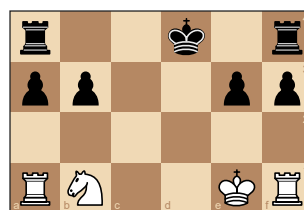
Files run `a`, `b`, ... and ranks `1`, `2`, ...; `a1` is the dark square in the lower-left corner, `h8` the upper-right. Square names are case-insensitive (`"E4"` = `"e4"`).

```
#board(
  "8/8/8/8/8/8/8/8",
  label-mode: "outside",
  size: 4cm,
)
```



The board is **not** tied to 8×8. A `position` built from the string form (next chapter) counts its own columns and rows, and the renderer draws whatever geometry it is given — files and ranks extend as far as the board needs, and the cells stay square while the board itself becomes rectangular:

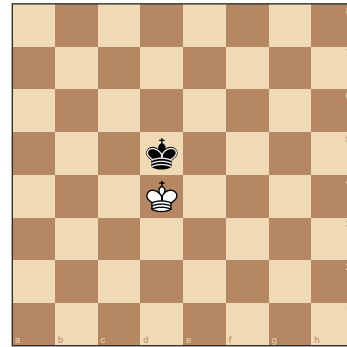
```
#board(
  position(
    "r..k.r",
    "pp..pp",
    ".....",
    "RN..KR",
  ),
  size: 4cm,
)
```



2.5 Sizing

The default board size suits an A4 two-column layout. A board reads the available space at its insertion point via `layout`, so it adapts to any column or page size without being told the geometry, and shrinks to fit if asked for more than fits. `size` may be a `length`, a `ratio` of the available width, or `auto`:

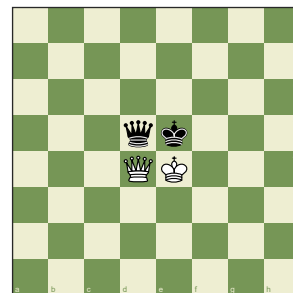
```
#board(
  "8/8/8/3k4/3K4/8/8/8",
  size: 60%, // of the available
width
)
```



2.6 Colours

`light` and `dark` set the two square colours:

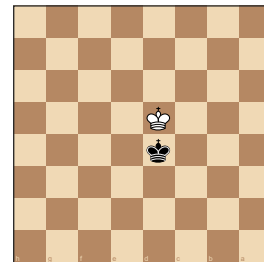
```
#board(
  "8/8/8/3qk3/3QK3/8/8/8",
  light: rgb("#eeeeed2"),
  dark: rgb("#769656"),
  size: 3.8cm,
)
```



2.7 Flip

`flip: true` shows the board from Black's side; labels, highlights and arrows all flip with it. Orientation is a **per-board** choice — `flip` is the one setting that cannot be made a document default (see **Document-wide style**).

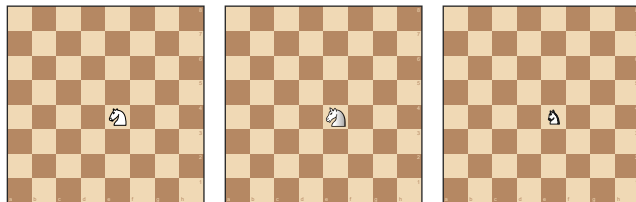
```
#board("8/8/8/3k4/3K4/8/8/8", flip:
true, size: 3.4cm)
```



2.8 Piece sets and fonts

Pieces are drawn from bundled **SVG piece sets**. Two ship with the package, both GPLv2+: `cburnett` (default) and `merida`. Choose one with `piece-set:` per board, or document-wide with `set-piece-set:`

```
#grid(columns: 3, gutter: 8pt, align: bottom,
  board("8/8/8/8/4N3/8/8/8", size: 2.6cm, piece-set: "cburnett"),
  board("8/8/8/8/4N3/8/8/8", size: 2.6cm, piece-set: "merida"),
  board("8/8/8/8/4N3/8/8/8", size: 2.6cm, piece-set: "unicode"),
)
```



The renderer accepts **any** set name and loads `src/assets/piece_sets/<name>/{w,b}{K,Q,R,B,N,P}.svg` on demand, so you can add a set by dropping such a folder into your copy and passing its name. `piece-set:` (or `none`) selects the glyph fallback `"unicode"`

— solid Unicode chess glyphs distinguished by fill and a contrasting stroke; it needs a font carrying them.

The rank/file **labels** are drawn in their own sans-serif, set by the `label-font` board option (a family or a fallback list), independent of the document font. The default is `("Arial", "DejaVu Sans Mono")` — Arial on Windows/macOS, falling back to Typst’s always-embedded mono — so a stock install draws labels without “unknown font family” warnings. Override it like any board default:

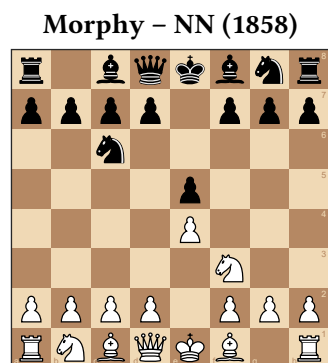
```
#set-board-defaults(label-font: "Segoe UI") // or a list, e.g. ("Helvetica",
"DejaVu Sans Mono")
```

3 Diagrams

`chess-diagram(source, ...)` wraps a board in a `#figure` (kind `"chess"`), so — unlike a bare `board` — it is captioned, counted, referenceable, and listed by an outline. `source` is the same FEN / position / squares the board takes, and it accepts every board option from the previous chapter.

A diagram carries two optional labels: a **game-info** line above the board (drawn automatically as `"<White> — <Black> (<Year>)"` when both players are known) and the figure **caption** below it.

```
#chess-diagram(
  "r1bqkbnr/pppp1ppp/2n5/4p3/4P3/5N2/
  PPPP1PPP/RNBQKB1R",
  white: "Morphy", black: "NN", year:
  1858,
  caption: [After 2...Nc6],
  size: 4.2cm,
)
```



`chess-diagram` is the standard-chess sugar over the generic `diagram`; both take the same source and overrides as `board`. Extra named arguments are forwarded to `figure` (e.g. `placement: top`).

3.1 Boards are not figures

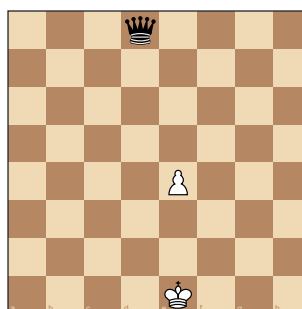
A bare `board` is plain content: it has **no** caption, **no** figure counter, does **not** resolve `@`-references, and is **not** listed by `chess-diagram-outline`. Only a **diagram** is a figure. So draw a `board` for an inline or

decorative position, and a `chess-diagram` whenever you want to caption it, cross-reference it (`@label`), or list it — see **Outlines and references**.

4 Positions

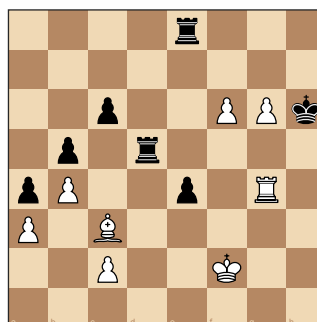
`position(...)` builds the data model — “which piece stands on which square” — that a board or diagram draws. It accepts a FEN string (auto-detected), a **squares dict**, or a **string form**. In a squares dict the piece can be a long name, a kind abbreviation, or a bare letter (UPPER = white, lower = black):

```
#chess-diagram(
  position((
    e1: (kind: "king", color:
"white"), // long
    d8: (kind: "q", color:
"black"), // kind
    e4:
"P", //
letter
  )),
  size: 4cm,
)
```



The **string form** reads like the board itself — first line is the TOP rank, `.` is empty. It is rectangular-only (this is what lets a board be non-8×8) and rejects characters that aren't a valid piece abbreviation or `.`:

```
#chess-diagram(
  position(
    "...r...",
    ".....",
    "..p..PPk",
    ".p.r....",
    "pP..p.R.",
    "P.B.....",
    "..P..K..",
    ".....",
  ),
  size: 4.2cm,
)
```



`position` returns a dict (`variant, cols, rows, squares, turn, castling, ; parse-fen` returns `en-passant, halfmove, fullmove`) the same shape. The `cols` / `rows` are counted from the string form (otherwise the 8×8 default).

5 Games (PGN)

`parse-pgn` returns an **array of games** (a PGN file may hold many); `.first()` takes one. Read an external file with `read` in your own file, or pass an inline raw block. The examples below assume a parsed game is in scope:

```
#let game = parse-pgn(read("game.pgn")).first()
```

Parsing is **lazy**: the roster (`tags`), the `result`, and the verbatim `movetext-row` are extracted cheaply; the move tree is built on demand by `movetext(game)`; the engine runs only when you ask for a position. So a tournament file read only for results never tokenises movetext.

`chess-notation(game)` renders the moves the game already holds, and `board-after(game, loc)` draws the position at a locator:

```
#chess-notation(game)
```

```
1. e4 e5 2. Nf3 Nc6 3. Bb5 a6 4. Ba4 Nf6 5. O-O Be7 6. Re1 b5 7. Bb3 d6 8. c3 O-O
```

```
#board-after(game, "3w", size: 4cm)
```

Alice – Bob (2024)



Diagram 5: Position after move 3. Bb5

5.1 Locators

A **locator** addresses one position in a game. The simple form is a string — `"30w"` / `"30b"`, the position after White's / Black's 30th **mainline** move. `position-after(game, loc)`, `board-after(game, loc, ..)`, and `to-fen(game, locator: ..)` all take it.

To address a move **inside a variation** (a PGN RAV), pass a **path** dict instead: `(line: (..hops..), at: "<final move>")`. Each hop is `(at: "<move>", into: <n>)` — branch off at mainline move `at`, descending `into` that move's variation number `n` (**0-based**: `0` is the first variation recorded at that move, `1` the second, ...). The top-level `at` is where you stop within the line you reached:

```
#let g = parse-pgn(
  "[White \"V\"] [Black \"T\"] 1. e4 (1.
  d4 d5 2. c4) e5 *",
).first()
// into variation 0 at White's move 1
// (the
// 1.d4 line), position after 2.c4:
#board-after(
  g,
  (line: ((at: "1w", into: 0)), at:
  "2w"),
  size: 3.4cm,
)
```

V – T

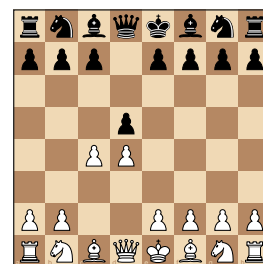


Diagram 6: Position after move 2. c4

Two easy traps:

- **The trailing comma.** A single-hop path is written `((at: "1w", into: 0),)` — the comma makes it a one-element **array** of hops. Without it, Typst reads a lone parenthesised dict and the locator is malformed.
- **Mainline is the fast path.** The string form (equivalently an empty `line: ()`) indexes a memoised list of mainline positions, so many diagrams off one game stay cheap; the path form is walked move by move.

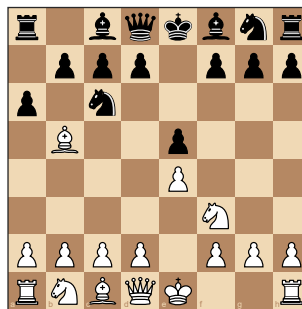
Addressing a variation that isn't there (`into`: past the recorded count) or a move past the end of its line is a hard error.

5.2 Playing moves onto a position

To explore a **new** line, or build a position from a FEN plus some moves, use `play-moves(source, moves)`. `source` is `none` (the standard start), a FEN string, or a position; `moves` is move text or a SAN array. It

resolves each move against the legal moves (illegal/ambiguous is a hard error) and returns the **final** position, never mutating the source:

```
#chess-diagram(  
  play-moves(none, "1. e4 e5 2. Nf3 Nc6  
3. Bb5 a6"),  
  size: 4cm,  
)
```



5.3 Notation output

`chess-notation(source, ..)` formats move text the game already holds — no engine. `source` is a game, a move-text string, or a SAN array. It can localise the piece letters and render figurine glyphs:

```
#chess-notation(game, lang: "de")
```

1. e4 e5 2. Sf3 Sc6 3. Lb5 a6 4. La4 Sf6 5. O-O Le7 6. Te1 b5 7. Lb3 d6 8. c3 O-O

```
#chess-notation(game, figurine: true)
```

1. e4 e5 2. ♘f3 ♙c6 3. ♗b5 a6 4. ♗a4 ♘f6 5. O-O ♗e7 6. ♖e1 b5 7. ♗b3 d6 8. c3 O-O

`from` / `to` restrict output to an **inclusive** slice of moves. Both are the simple `"8b"` / `"12w"` locators; `from` defaults to the first move, `to` to the last:

```
#chess-notation(game, from: "2w", to: "3b")
```

2. Nf3 Nc6 3. Bb5 a6

Unlike `board-after`, these are **mainline-only** — they do not accept the variation **path** form, and a `from` past the end or a `to` before `from` is a hard error.

Other options: `move-numbers`, `result`, and — for a **game** source — `nags` / `comments` (consulting the PGN-handling defaults). Localization substitutes only the piece letters; files, ranks, captures, check marks and `0-0` are untouched.

5.3.1 Programmatic NAGs

`nags: true` renders the NAGs a game already carries (the `$n` glyphs in the PGN). To attach NAGs **without editing the PGN**, use `with-nags(game, map)`: it returns a **new** game whose addressed mainline moves carry the given NAGs, which `notation(..., nags: true)` then renders.

```
#let g = parse-pgn(  
  "[White \"A\"][Black \"B\"] 1. e4 e5 2. Nf3 Nc6 *",  
)  
#chess-notation(  
  with-nags(g, ("2w": "!", "2b": "$14")),  
  nags: true,  
)
```

1. e4 e5 2. Nf3! Nc6±

Each map key is a **mainline** locator ("2w" / "2b"); each value is a "\$n" code, one of the six suffix glyphs ! ? !! ?? !? ?! (sugar for \$1 – \$6), or an array of those. The mapping **replaces** any NAG already on that move, and the source game is never mutated. Only mainline moves are addressable — notation renders the mainline only.

5.4 Exporting FEN

to-fen is the inverse of parse-fen . It serialises a position, or a game at a locator. Standard 8×8 positions round-trip exactly:

```
#raw(to-fen(play-moves(none, "1. e4 e5 2. Nf3")))
```

```
rnbqkbnr/pppp1ppp/8/4p3/4P3/5N2/PPPP1PPP/RNBQKB1R b KQkq - 1 2
```

5.5 Drawing annotations

PGN comments can carry drawing annotations — [%cal ...] for arrows, [%csl ...] for highlights. Processing is **off by default**; opt in per call with annotations: true (or document-wide with set-pgn-defaults). The demo game annotates its 2nd move:

```
// move 2: {%cal Gf3e5} [%csl Re5]}
#board-after(game, "2w", annotations:
true, size: 4cm)
```

Alice – Bob (2024)



Diagram 8: Position after move 2. Nf3

The color letters (G R Y B O) resolve through the annotation-colors board style; annotations merge with any arrows / highlight you pass explicitly.

With embedded diagrams. The diagrams switch (PGN handling) makes notation(diagrams: true) splice a board after each move whose comment carries a diagram marker. If that same comment also holds %cal / %csl and annotations is on, the spliced board shows those arrows/highlights too — the two switches compose:

```
// 2. Nf3 {%cal Gf1c4} [%csl Re5} #[After 2.Nf3]} Nc6 ...
#set-pgn-defaults(diagrams: true, annotations: true)
#chess-notation(game) // ...text, then a board after 2.Nf3 with the arrow +
highlight
```

diagrams decides **whether** a board appears; annotations decides whether it carries the marks. The marker and the %cal / %csl must be in the **same** move's comment. (Only mainline moves are addressed — notation renders the mainline.)

5.6 Errors

Malformed PGN is a **hard error**: broken tag syntax and stray variation parens fail at parse time; illegal, ambiguous, or unparseable moves fail when the position is navigated. Missing Seven-Tag-Roster tags are tolerated (they default).

6 Tournament tables

Tournament tables are built from a parsed PGN's roster + results (no engine). The `*-table` renderers produce a captioned, referenceable `#figure` (kind `"chess-table"`). Each works on a list of games, with `by: "player"` or `by: "team"`. The examples use a parsed 4-player round-robin in `games`:

```
#let games = parse-pgn(read("event.pgn"))
```

`standings-table` sorts best-first (score, then tie-breaks, then first appearance):

```
#standings-table(games, by: "player")
```

Pos	Player	Pl	+	=	-	Pts	Bh	SB
1	Alice	3	2	1	0	2½	3½	2½
2	Dave	3	1	2	0	2	4	2½
3	Bob	3	1	0	2	1	5	½
4	Carol	3	0	1	2	½	5½	1

`crosstable-table` renders the round-robin grid (it **requires** a complete round-robin and errors otherwise — use `standings` + `progress` for Swiss/league):

```
#crosstable-table(games, by: "player")
```

#	Player	1	2	3	4	Pts
1	1 Alice		½	1	1	2½
2	2 Dave	½		1	½	2
3	3 Bob	0	0		1	1
4	4 Carol	0	½	0		½

`progress-table` shows the round-by-round running score (needs the `Round` tag):

```
#progress-table(games, by: "player")
```

#	Player	R1	R2	R3	Pts
1	Alice	1 (1)	1 (2)	½ (2½)	2½
2	Dave	½ (½)	1 (1½)	½ (2)	2
3	Bob	0 (0)	0 (0)	1 (1)	1
4	Carol	½ (½)	0 (½)	0 (½)	½

Each renderer takes `caption` (used by refs and the outline), `title` (a heading above the table), `supplement`, and `lang`. The compute functions (`standings`, `crosstable`, `progress`) are also public, returning plain data for custom layouts. **Team** mode groups games by the `Round = "round.board"` convention into matches.

7 Outlines and references

Diagrams and tables each carry a distinct figure `kind` (`"chess"` / `"chess-table"`), so they get their own counters, can be **referenced** (`@label` → "Diagram 3" / "Table 2"), and can be **listed separately**:

```
#chess-diagram-outline() // list of chess diagrams
#chess-table-outline()  // list of tournament tables
```

```
#chess-outlines() // both, diagrams then tables

#chess-diagram(starting-fen, caption: [Start]) <start>
As shown in @start, ...
```

Only figures that carry a **caption** appear in an outline (a caption-less figure is still referenceable but unlisted, matching Typst). Titles are language-aware by default and settable per call (`title:` , `lang:`) or document-wide (`set-diagram-defaults(outline-title: ..)`). Remember that a bare `board` is not a figure, so it can be neither referenced nor listed — use a `chess-diagram` .

8 Document-wide style

Defaults live in **five** buckets, each with its own setter:

bucket	setter	controls
board	set-board-defaults	square colours, labels, piece set, highlights, arrows, grid, size
diagram	set-diagram-defaults	the diagram figure: game-info line, supplement, outline title
table	set-table-defaults	the table figure: supplement, outline title, title gap
language	set-lang	the document language (also set-il8n-defaults)
PGN handling	set-pgn-defaults	annotations / nags / comments / diagrams

`set-chess-defaults` is the single **umbrella** over **all five** buckets: pass any key from any bucket — **including** `lang` — and it is routed to the bucket that owns it.

```
#set-board-defaults(light: rgb("#eeeeed2"), dark: rgb("#769656"), size: 5cm)
#set-lang("de")
#set-pgn-defaults(nags: true)
// ...or all of the above at once, through the umbrella:
#set-chess-defaults(dark: blue, lang: "de", nags: true, info-gap: 1em)
#set-piece-set("merida") // sugar for set-board-defaults(piece-set: ..)
```

A setter affects **every subsequent** diagram/table; per-call arguments still override. Every default is equivalently a per-call argument — the same green theme, set once above vs. passed to one diagram:

```
#chess-diagram(
  "rnbqkbnr/pppppppp/8/8/8/PPPPPPPP/
RNBQKBNR",
  light: rgb("#eeeeed2"), dark:
  rgb("#769656"),
  size: 4cm,
)
```

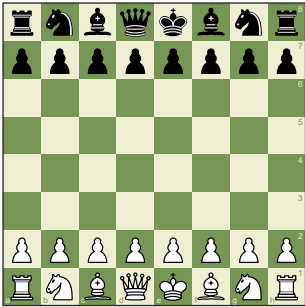


Diagram 9: Position at move 1, White to play

Two caveats. `flip` is **not** allowed in any defaults setter — orientation is a per-board choice, so `set-chess-defaults(flip: ..)` is an error. And `supplement` / `outline-title` live in **both** the diagram and table buckets; the umbrella routes them to **diagram**, so use `set-table-defaults` for the table ones.

8.1 Language

A single document **language** drives every language-aware string — diagram and table supplements, outline titles, and notation piece letters. Default is English; "auto" follows `#set text(lang: ..)`; or pick a code:

```
#set-lang("de")      // Diagramm / Tabelle / Sf3, Lb5, ...
#set-lang("auto")    // follow #set text(lang: ..)
```

Every localizable string is also per-call overridable (the `lang:` argument seen on `chess-notation` above). Adding a language is a no-code change: drop a `src/assets/il8n/<code>.typ` and register it in `src/il8n.typ`.

8.2 PGN handling

A **PGN-handling** bucket decides how much of a parsed game's embedded extras get interpreted at render time. Parsing stays lossless; these switches only decide what is processed, and **all default off**:

```
#set-pgn-defaults(annotations: true, nags: true, comments: true)
```

key	effect
annotations	<code>%cal / %csl</code> → arrows/highlights on <code>board-after</code>
nags	render NAGs (<code>Nf3!</code> , <code>d4±</code>) in <code>notation</code>
comments	include comment prose in <code>notation</code>
diagrams	embed a board in <code>notation</code> after each move marked for one

Each is also a per-call argument (`auto` → the document default) on `notation` / `board-after` — as used in the annotations example above.